

ASTRAKHANKA, Russian Federation

WMO: 319210

Lat: 44.7169N

Lon: 132.0730E

Elev: 256

StdP: 14.56

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.7	-10.4	-22.1	1.6	-12.0	-18.4	2.0	-7.6	22.4	12.6	17.8	11.7	3.5	220	0.561

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.7	84.1	73.5	81.4	71.9	78.9	70.6	75.7	81.5	74.0	79.2	72.3	76.9	7.1	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.8	127.3	79.3	72.2	120.3	77.6	70.6	113.8	75.7	39.5	81.8	37.9	79.4	36.3	77.0	91.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.6	17.2	14.8	DB	-20.0	90.3	6.2	2.3	-24.4	92.0	-28.0	93.3	-31.5	94.6	-35.9	96.2	(4)
(5)			WB	-20.3	79.1	6.1	3.4	-24.6	81.5	-28.2	83.5	-31.6	85.4	-36.0	87.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	40.5	3.3	11.6	26.2	42.4	55.1	64.1	70.6	70.1	59.9	45.4	26.2	8.6		
	DBStd	24.64	8.32	9.04	9.11	7.31	6.10	5.10	4.32	4.47	6.08	7.88	10.79	8.57		
	HDD50	5715	1448	1075	738	248	20	0	0	0	4	186	714	1282		
	HDD65	9382	1913	1495	1203	677	311	76	6	10	173	608	1163	1747		
	CDD50	2231	0	0	0	21	179	424	639	623	301	43	1	0		
	CDD65	422	0	0	0	0	5	49	181	167	20	0	0	0		
	CDH74	2188	0	0	0	8	83	308	962	744	81	2	0	0		
	CDH80	392	0	0	0	0	16	56	203	110	7	0	0	0		
(14) Wind	WSAvg	7.2	5.4	6.4	7.5	8.6	8.8	8.1	7.1	6.6	6.8	7.4	7.2	6.0		
(15) Precipitation	PrecAvg	22.9	0.4	0.4	0.7	1.4	2.8	2.8	4.2	4.4	2.9	1.9	0.8	0.6		
	PrecMax	30.1	1.4	1.1	2.2	3.0	5.3	5.5	8.5	8.4	7.1	5.0	1.7	1.4		
	PrecMin	15.1	0.0	0.0	0.0	0.3	0.8	0.6	0.8	0.4	0.4	0.6	0.2	0.1		
	PrecStd	4.2	0.4	0.3	0.5	0.8	1.3	1.2	1.9	2.0	1.8	1.1	0.5	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.7	40.1	56.1	74.0	82.1	85.8	88.3	86.8	80.2	70.8	57.3	38.2		
		MCWB	27.7	34.3	44.1	55.4	63.8	71.4	76.8	75.1	68.6	58.1	47.6	35.1		
	2%	DB	24.6	34.6	46.8	65.1	75.3	80.7	84.9	82.7	76.1	65.2	50.8	31.3		
		MCWB	22.0	31.2	38.4	50.6	60.0	69.2	75.1	73.3	66.5	55.7	44.2	28.7		
	5%	DB	19.9	30.9	41.3	59.0	70.5	77.3	82.0	80.3	73.0	61.0	45.5	26.1		
		MCWB	17.4	27.5	35.4	47.9	58.1	67.8	73.3	72.5	64.2	53.5	40.4	23.5		
	10%	DB	16.0	26.0	38.0	54.1	66.4	74.3	79.2	78.1	70.0	57.7	40.9	21.7		
		MCWB	13.8	23.0	33.4	45.6	56.6	66.0	71.8	71.0	63.0	51.9	36.6	19.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.4	35.3	44.8	56.8	66.0	74.4	79.7	77.5	71.9	61.0	50.0	35.8		
		MCDB	29.9	38.4	54.9	70.0	78.8	81.5	85.4	83.7	76.5	66.8	54.7	37.5		
	2%	WB	22.0	31.6	39.4	52.3	62.1	71.1	76.4	75.5	68.5	57.9	44.9	28.8		
		MCDB	24.7	34.3	45.5	62.5	72.0	77.9	82.7	80.8	73.5	62.6	50.0	31.0		
	5%	WB	17.6	27.5	36.3	49.4	59.5	68.7	74.5	74.0	66.4	55.6	40.9	23.5		
		MCDB	20.0	30.7	40.4	57.5	68.1	75.7	80.3	78.5	70.9	59.9	44.7	25.7		
	10%	WB	13.9	23.1	34.1	46.5	57.3	66.7	72.6	72.5	64.4	52.5	36.8	19.2		
		MCDB	16.0	26.0	37.5	53.6	64.7	73.3	77.9	76.4	68.8	56.7	40.5	21.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.6	17.6	15.0	15.6	16.5	15.8	13.7	13.1	16.6	17.4	15.5	15.2		
		MCDBR	20.2	20.7	19.5	26.1	26.4	22.1	18.9	16.4	20.2	22.7	21.9	19.5		
	5% WB	MCWBR	17.8	17.6	13.2	14.8	13.5	12.3	10.2	8.8	11.3	14.2	16.2	17.2		
		MCDWR	20.1	19.9	18.4	23.9	22.3	19.7	17.0	14.2	16.0	18.8	20.8	19.1		
(40) Clear-Sky Solar Irradiance	taub		0.321	0.371	0.456	0.567	0.582	0.563	0.522	0.482	0.437	0.459	0.389	0.336		
	taud		2.057	1.920	1.829	1.694	1.730	1.863	2.026	2.133	2.158	1.952	2.006	2.021		
	Ebn at Noon		239	246	239	224	226	232	240	246	246	213	208	216		
	Edh at Noon		33	46	58	72	71	63	53	46	41	44	35	31		
(44) All-Sky Solar Radiation	RadAvg		691	1039	1383	1468	1646	1768	1582	1489	1333	948	647	554		
	RadStd		46	48	104	134	135	188	155	127	108	71	50	29		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (16 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page